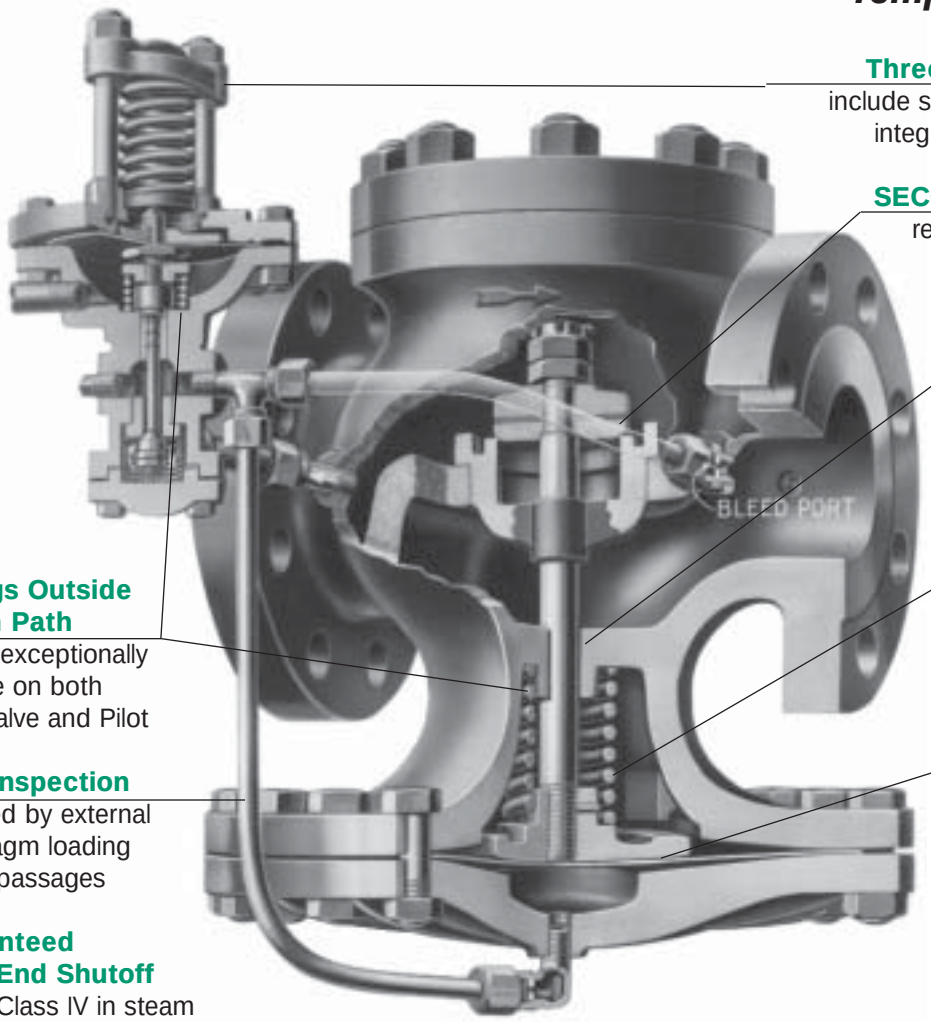
Applications

- Pressure Regulation for Steam Distribution
- Single Point or Multiple Use Applications
- Pressure Control for Steam Plants
- District Heating Systems
- Single Stage Reduction Stations
- Two Stage Reduction Stations
- Parallel Reduction Stations

Iron Horse ED Series Pressure Regulator

Pressures To 600 PSIG
Temperatures to 750°F



Springs Outside Steam Path
assure exceptionally long life on both Main Valve and Pilot

Easy Inspection
provided by external diaphragm loading steam passages

Guaranteed Dead End Shutoff
meets Class IV in steam service, even on large sizes

Patented *SECOWELD* Option
allows easy repair of seat ring threads damaged by high pressure applications

Three Pilot Mounting Options
include standard side mount (shown), integral mount and remote mount

SECO Metal Seats and Discs
resist wiredraw - not one case of SECO Metal being cut by steam in 75 years

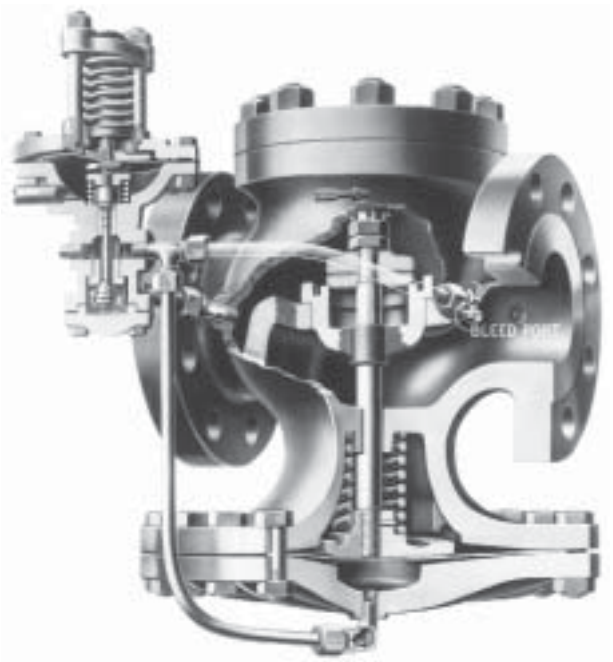
Packless Construction
eliminates leakage and greatly reduces friction and stem wear

Two Main Spring Options
for superior regulation over a wide range of applications

Large, Protected Metal Diaphragm
bathed in condensate, sealed away from steam seldom needs replacement

No Closely Fitted Parts
to stick or bind due to uneven expansion or foreign matter

Few Moving Parts
mean long service life



TYPE ED SERIES
PRESSURE REGULATOR

CAST IRON or STEEL
PRESSURES to 600 PSIG at 750°F

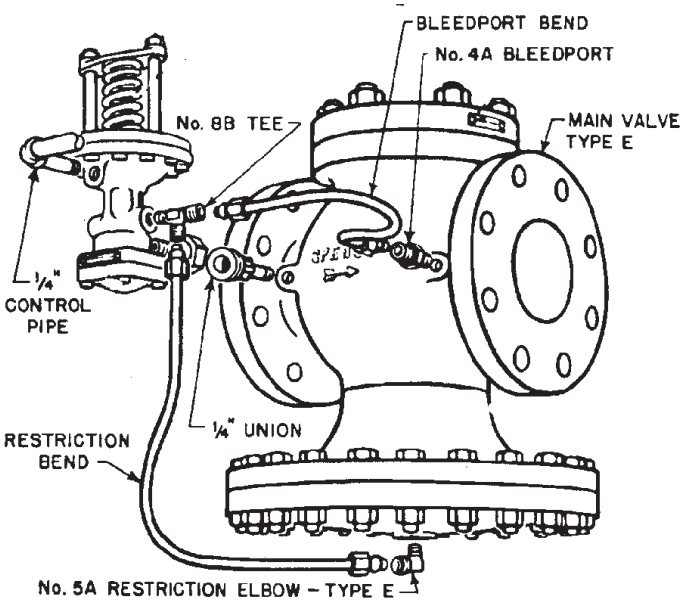
APPLICATION DATA

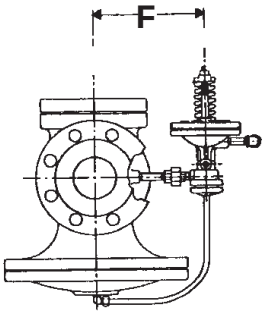
- **Pressure Regulating for Steam Distribution**
- **Single Point or Multiple use Applications**
- **Pressure Control for Steam Plants**
- **District Heating Systems**
- **Single Stage Reductions**
- **Two Stage Reductions**
- **Parallel Reduction**

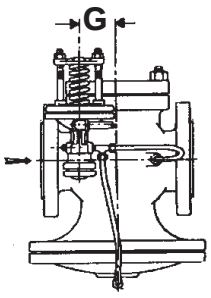
TYPE ED PRESSURE REGULATOR

VALVE INFO
PAGE 28

PILOT INFO
PAGE 48







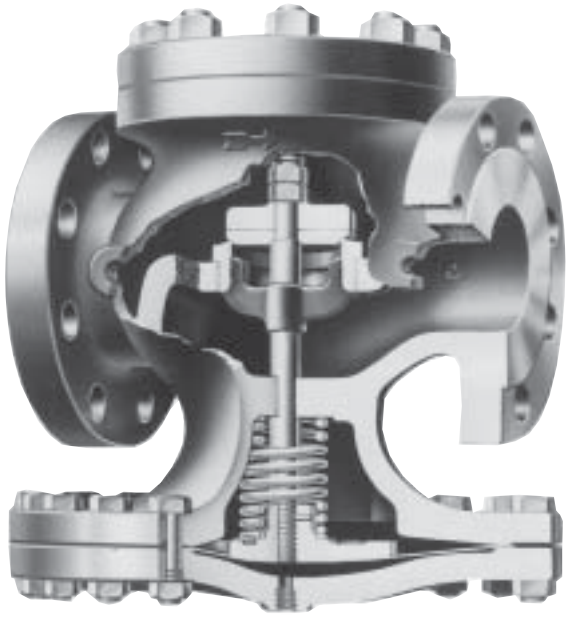
Valve is tapped so that Pilot may be mounted on either side.

DIMENSIONS
inches (mm)

SIZE	F	G
3/8 (10)	5 3/8 (136)	1 1/4 (32)
1/2 (15)	5 3/8 (136)	1 1/4 (32)
3/4 (20)	5 3/8 (136)	1 3/8 (35)
1 (25)	5 3/4 (146)	1 1/2 (38)
1 1/4 (32)	6 (152	1 7/8 (48)
1 1/2 (40)	6 1/4 (159)	2 (51)
2 (60)	6 5/8 (168)	2 1/8 (54)
2 1/2 (65)	6 3/4 (171)	2 3/8 (60)
3 (80)	7 1/4 (184)	2 3/4 (70)
4 (100)	8 (203)	3 1/2 (89)
5 (125)	9 (229)	3 1/2 (89)
6 (150)	9 7/8 (251)	4 (102)
8 (200)	10 1/2 (267)	6 1/4 (159)
10 (250)	12 1/2 (318)	6 (152)
12 (300)	14 (356)	8 1/2 (216)

TYPE E MAIN VALVE

SIZES 3/8" – 12"
PRESSURES to 600 PSIG at 750°F



TYPE E MAIN VALVE

- **Normally Closed**
- **Single Seat**
- **Balanced Metal Diaphragms**
- **Protected Main Spring**
- **Fluid, Gas & Vapor Applications**
- **Multiple Trims for Precise Sizing**
- **ANSI/FCI 70-2 Class IV Shutoff**
- **Virtually Frictionless for Long Service Life**
- **Packless Construction**
- **Easy In-line Maintenance**
- **Wide Variety of Pilots for Many Applications**
- **Minimum Operating ΔP 10 psi (.7 bar)**
- **Lifetime Warranty against Wiredrawing of Seat & Disc***

Application Data

- Pressure Regulating for Steam Distribution
- Regulating for Process Control (Temperature or Pressure)
- Maintain Back Pressure or Differential Pressure
- For use with Self-contained, Pneumatic or Electronic Pilots
- Single Point or Multiple Use Applications
- Slow Start-up or Shutdown

SIZING INFO
PAGE 108

Valve Ratings

Valve Ends ASME/ANSI	Pressure PSIG (bar)		Temperature °F (°C)
CAST IRON			
Class 250 NPT	250 (17.2)	@	450 (232)
B16.1 Class 125 Flanged	125 (8.6)	@	450 (232)
B16.1 Class 250 Flanged	250 (17.2)	@	450 (232)
CAST STEEL			
B16.34 Class 300 NPT	300 (21.0)	@	600 (315)†
B16.34 Class 150 Flanged	150 (10.3)	@	500 (260)
B16.34 Class 300 Flanged	300 (21.0)	@	600 (315)†
B16.34 Class 600 Flanged	600 (41.4)	@	600 (315)†

†750°F (400°C) construction available on request.
Other pressure/temperature ratings available; consult factory.
Maximum downstream pressure is 300 psi.

Canadian Registration # OC 0591.9C

Installation Tip: Add EZ Connections for ease of maintenance
SEE PAGE 42

Options

- Composition Disc *See page 42*
- Balanced Construction
- Insulcap Insulating Jacket
- High Temperature Construction
- Low ΔP (LP) Main Spring
- Parabolic Disc
- Integral Mount Pilot
- Secoweld
- Dashpot
- EZ Connections

Typical Configurations

Pressure Reducing .....Type ED Series
Air Adjusted .....Type EA Series
Back Pressure .....Type EQ Series
Pump Governor .....Type EP Series
Load Allocating .....Type EFD
Air Controlled .....Type EAP60
Electronic Slow Start .....Type ED208D
Solenoid Controlled .....Type EMD
Solenoid Actuated .....Type EM
Differential .....Type EN
Temperature Control .....Type ET Series

RATED FLOW COEFFICIENTS (Cv)

SEAT FACTOR	REGULATOR SIZE														
	3/8	1/2	3/4	1	1 1/4	1 1/2	2	2 1/2	3	4	5	6	8	10	12
Full	1.5	2.8	5.4	8.8	14.1	19.8	31	44	74	109	169	248	444	706	1113
Full 75 %	—	2.1	4.0	6.6	10.6	14.8	23.3	33	56	82	127	186	333	530	835
Full 50 %	—	1.4	2.7	4.4	7.0	9.9	15.5	22	37	55	85	124	222	353	557
Normal	.65	1.5	4.8	7.5	10.4	14.6	17.6	24	43	78	115	151	249	377	631
Normal 75 %	—	—	—	—	—	—	—	18	33	59	87	114	187	283	474
Normal 50 %	—	—	—	—	—	—	—	12	22	39	58	76	125	189	316

* When installed according to factory specifications.

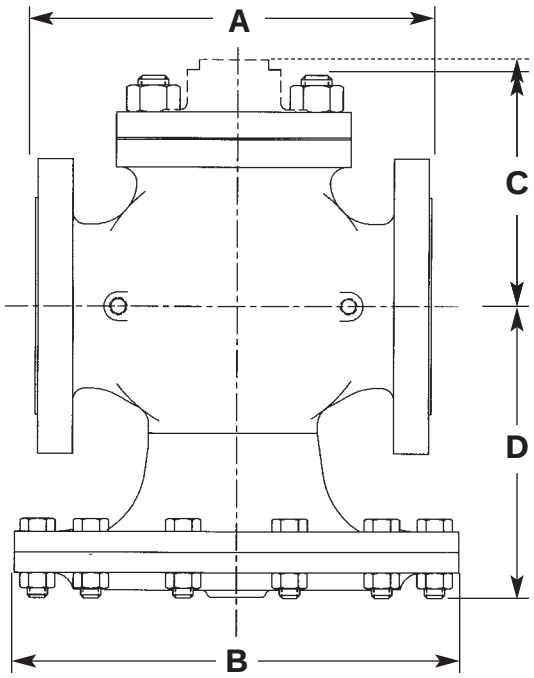
TYPE E MAIN VALVE

SPECIFICATION

The valve shall be self-operated, external pilot type, single seated, metal diaphragm actuated, normally closed design. The valve will function quickly and shut tight on dead end service. Internal parts including seats, discs, stems and diaphragms shall be of stainless steel. There shall be no springs in the steam space and no stuffing box. The valve shall be easy to maintain with all parts accessible without removal from the line.

MATERIALS OF CONSTRUCTION

Body, Cast Iron .....ASTM A126 Cl. B
Body, Cast Bronze .....ASTM B61 UNS C92200
Body, Cast Steel .....ASTM A216 WCB
Stem .....303 St. Stl. ASTM A582
Disc 3/4 - 5" .....420 St. Stl. ASTM A743 CA-40
Disc 6 - 12" .....304 St. Stl. ASTM A167/A240
Seat 3/4 - 5" .....420 St. Stl. ASTM A743 CA-40
Seat 6 - 12" .....316 St. Stl. ASTM A743-79 CF-8M
Gasket .....Non-asbestos
Diaphragm .....Stainless Steel MIL-S-5059C
Spring .....Steel



TYPE E MAIN VALVE

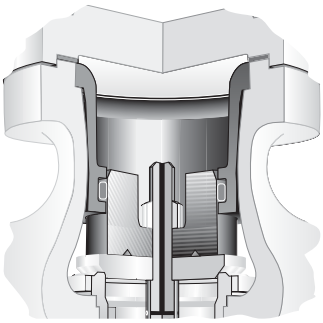
FITTINGS ON PAGE 46

DIMENSIONS inches (mm) **AND WEIGHTS** pounds (kg)

SIZE	FACE TO FACE DIMENSIONS A ANSI NPT	A ANSI 125,150	A ANSI 250	A ANSI 300	A ANSI 600	B	C Std. Mount ANSI 600	C Integral Mount ANSI 600	C Integral Mount Cl & Brz. All	C Integral Mount Steel	C Integral Mount Steel 600	D*	APPROX. WT. ANSI NPT	APPROX. WT. ANSI 125,150	APPROX. WT. ANSI 250,300	APPROX. WT. ANSI 600
3/8 (10)	4 3/8 (111)	— —	— —	— —	— —	5 7/8 (149)	2 3/4 (70)	— —	3 1/2 (89)	3 1/2 (89)	— —	5 1/4 (133)	14 (6)	— —	— —	— —
1/2 (12)	4 3/8 (111)	— —	— —	— —	6 (152)	5 7/8 (149)	2 3/4 (70)	2 3/4 (70)	3 1/2 (89)	3 1/2 (89)	3 5/8 (92)	5 1/4 (133)	14 (6)	— —	— —	20 (9.1)
3/4 (19)	4 3/8 (111)	— —	— —	— —	6 3/8 (162)	6 1/2 (165)	2 7/8 (73)	3 7/8 (98)	3 5/8 (92)	3 3/4 (95)	4 1/2 (114)	5 1/2 (140)	18 (8)	— —	— —	28 (13)
1 (25)	5 3/8 (137)	5 1/2 (140)	6 (152)	6 1/2 (165)	6 1/2 (165)	7 (178)	3 5/8 (92)	4 1/4 (108)	4 3/8 (111)	4 3/8 (111)	4 3/4 (121)	6 1/4 (159)	23 (10)	26 (12)	31 (14)	32 (15)
1 1/4 (32)	6 1/2 (165)	6 3/4 (171)	7 1/4 (184)	7 7/8 (200)	7 7/8 (200)	7 7/8 (200)	4 1/8 (105)	4 5/8 (117)	4 (102)	4 5/8 (117)	5 (127)	6 1/2 (165)	33 (15)	37 (17)	41 (19)	45 (20)
1 1/2 (38)	7 1/4 (184)	6 7/8 (175)	7 3/8 (187)	8 (203)	8 (203)	8 3/4 (222)	4 3/8 (111)	5 1/8 (130)	4 3/8 (111)	5 (127)	— —	7 1/8 (181)	43 (20)	47 (21)	55 (25)	58 (26)
2 (51)	7 1/2 (191)	8 1/2 (216)	9 (229)	10 1/4 (260)	10 1/4 (260)	9 7/8 (251)	5 1/4 (133)	5 3/4 (146)	5 (127)	5 5/8 (143)	5 3/4 (146)	7 5/8 (194)	62 (28)	73 (33)	78 (35)	83 (38)
2 1/2 (64)	— —	9 3/8 (238)	10 (254)	11 1/4 (286)	11 1/4 (286)	10 7/8 (276)	5 3/4 (146)	7 7/8 (200)	5 1/2 (140)	6 (152)	8 1/4 (210)	8 3/8 (213)	— —	95 (43)	100 (45)	130 (59)
3 (76)	— —	10 (254)	10 3/4 (273)	12 1/4 (311)	12 1/4 (311)	11 3/4 (298)	6 5/8 (168)	9 1/8 (232)	6 3/8 (162)	7 1/8 (181)	— —	9 1/4 (235)	— —	125 (57)	140 (64)	175 (80)
4 (102)	— —	11 7/8 (302)	12 1/2 (318)	12 1/2 (318)	14 1/2 (368)	14 3/4 (375)	7 5/8 (194)	10 5/8 (270)	7 1/4 (184)	8 (203)	— —	11 7/8 (302)	— —	210 (95)	230 (105)	310 (141)
5 (127)	— —	13 5/8 (346)	14 1/2 (368)	14 1/2 (368)	16 1/2 (419)	16 7/8 (429)	8 1/2 (216)	12 1/2 (318)	8 1/8 (206)	8 1/2 (216)	— —	12 1/2 (318)	— —	295 (134)	310 (141)	490 (223)
6 (152)	— —	15 1/8 (384)	16 (406)	16 (406)	17 3/8 (441)	19 3/4 (502)	10 (254)	13 3/4 (349)	9 1/2 (241	9 1/2 (241)	13 5/8 (346)	14 1/8 (359)	— —	420 (191)	470 (214)	655 (298)
8 (203)	— —	19 (483)	20 (508)	20 (508)	21 5/8 (549)	22 1/2 (572)	11 1/2 (292)	15 3/8 (391)	11 1/4 (286)	11 3/4 (298)	— —	17 1/4 (438)	— —	700 (318)	710 (323)	1070 (486)
10 (254)	— —	23 5/8 (600)	25 (635)	25 (635)	— —	28 (711)	13 3/4 (349)	— —	— —	— —	— —	23 3/8 (594)	— —	1240 (563)	1300 (591)	— —
12 (305)	— —	26 1/2 (673)	28 (711)	28 (711)	— —	33 (838)	15 7/8 (403)	— —	— —	— —	— —	25 1/4 (641)	— —	2060 (936)	2140 (972)	— —

*Add 65% to D dimension for stem removal clearance.

MAIN VALVE OPTIONS



Balanced Construction

There are installations where it is desirable to not have the inlet pressure forcing down on the Main Valve Disc. In these instances, the E Main Valve should be internally balanced. The balance parts allow the down-stream pressure to rest on top of the disc, thus allowing for finer adjustments in the Main Valve travel and a smoother operating regulator.

SECOWELD

The greatest weakness in a High Pressure Valve is the threaded joint between the Seat Ring and the body. A slight leak developing at this point will gradually erode the Body metal, thus accentuating the leak and eventually ruining the body. Various impractical schemes, such as welding the Seat Ring into the Body, have been tried to overcome this weakness. The invention SECOWELD solves this problem and, at the same time, provides an easily renewable Seat Ring. In the SECOWELD Design, a SECO Metal Bushing is welded to and thus sealed in the Body and, in turn, is threaded to take the Main Seat Ring, which is also of SECO Metal. As SECO Metal resists wiredrawing, if slight leakage should occur, no damage can be done to the body or to the threads of either SECO Metal piece.





EZ CONNECTIONS

Provides the performance of a flanged connection with the simplicity of a union connection. Unlike conventional unions, EZ Connections do not require matched sets or springing pipe to clear cone tolerances and do not leak after just a few disassembly/reassembly cycles. Uniform end to end dimensions simplify rough-in schematics. Available on ½" through 2" threaded main valves in NPT, socketweld and threaded by socketweld connections.

See Uniflex Couplings on page 374

Condensation Chamber

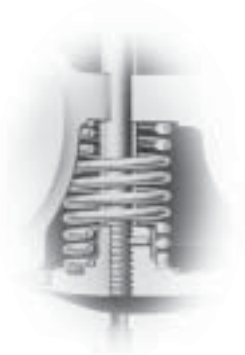
A Condensation Chamber is standard on the Type E5 Main Valve. A Condensation Chamber is standard on the Type E Main Valve when steam temperatures exceed 600°F. Any Main Valve discharging steam into a vacuum should include a Condensation Chamber.



MAIN VALVE OPTIONS

Low Differential Pressure (LP) Main Spring

The E Series Main Valves provide superior regulation in a broad range of applications by utilizing a specialized Main Spring. When differential pressures between 10-50 psi are desired, E Main Valves should be equipped with the optional LP Main Spring. The LP Main Spring alone will achieve differential pressures to 15 psi. In order to attain differential pressures to 10 psi, optional 5B Open Elbow and 1/16" 4A Bleedport are required.





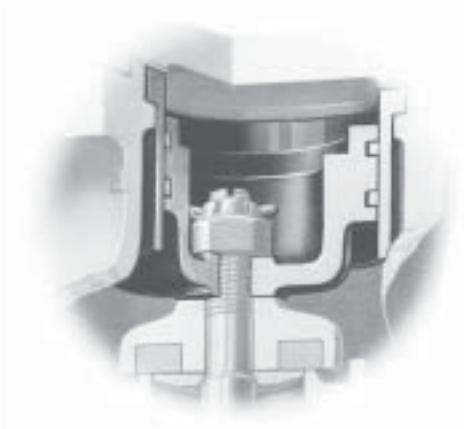
COMPOSITION DISC

In a Single Seat Main Valve, the Integral (all-metal) Disc is interchangeable with the Composition Disc Assembly. The Composition Disc is recommended for service on air, gas and water where absolutely tight shutoff is required and is available on Full and Normal seats and Parabolic valve plugs. The Composition Disc is suitable for pressures to 200 psi and temperatures to 200ºF.

PARABOLIC DISC

In order to meet special flow requirements, any Spence Main Valve can be equipped with a Parabolic or other specially shaped Disc. Due to the fact that the Spence Main Valve is operated by a large, balanced Diaphragm and is nearly frictionless in operation, special Discs are not required on normal installations.





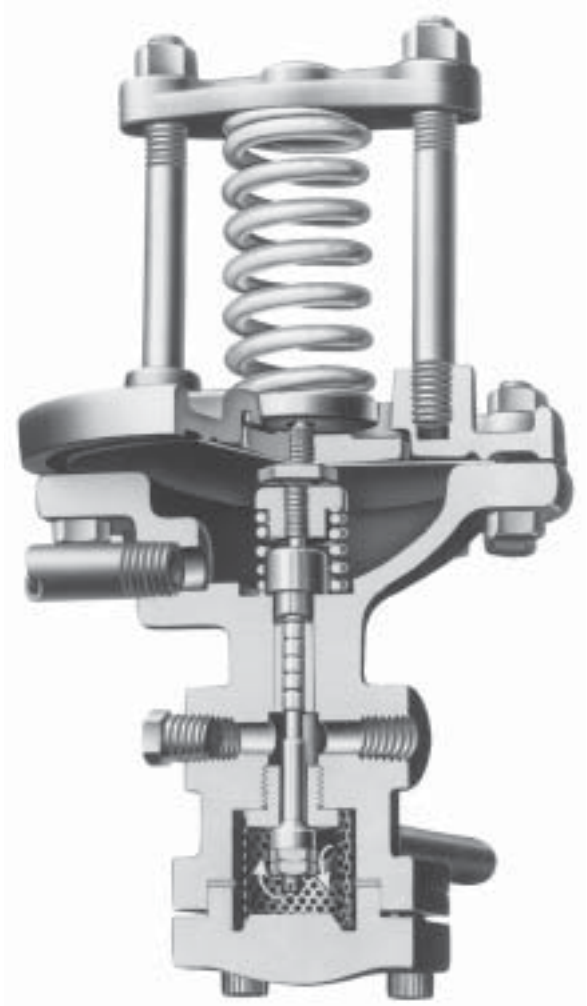
DASHPOT

In order to prevent water hammer, Dashpots are required in all single seat, normally closed Main Valves used on liquid service, except Type C34. Dashpots are neither necessary nor desirable on steam, air or gas service and are not required in double seat valves or in normally open single seat valves. Illustration shows Dashpot and Composition Disc for initial pressures of 200 psig and less. For initial pressures greater than 200 psig, standard metal to metal seat and disc are used.

TYPE D
PRESSURE REDUCING PILOT

CONTROLS 3 to 300 PSIG

- **Self Contained**
- **Spring Operated**
- **Normally Closed**
- **Packless Construction**
- **Fluid, Gas & Vapor Applications**
- **Accurate Regulation Unaffected by Service Conditions**
- **Easy In-line Maintenance**



TYPE D PRESSURE PILOT

Models

- **TYPE D** for ±1 psi control of delivery pressures between 3 and 150 psi.
- **TYPE D2** for control of delivery pressures between 100 and 300 psi.
- **TYPE D5** for ±1/2 psi control of delivery pressures between 1 and 25 psi.
- **TYPE D120** for exceptionally fast response controlling delivery pressures between 5 and 150 psi.

Options

- Spring Chamber
- Composition Disc
- Vacuum Spring Assembly
- Adjusting Handwheel
- Integral Mount Body

Typical Configurations

Pressure Reducing .....Type ED

Pressure Reducing .....Type E2D

Pressure Reducing .....Type E5D

Pressure Reducing .....Type E6D

Water Pressure Reducing.....Type C34D

Temperature & Pressure .....Type ET14D

Application Data

- Pressure Regulating for Steam Distribution
- Regulating for Process Control
- Can be used with Temperature Pilot to Regulate Pressure on Temperature control Application

SIZING INFO
PAGE 117

Ratings (Maximum Inlet Conditions)

Construction	Pressure PSIG (bar)		Temperature °F (°C)
Cast Iron	250 (17.2)	@	450 (232)
Cast Steel	600 (41.4)	@	750 (400)

Spring Pressure Ranges (PSIG)

TYPE D	TYPE D2	TYPE D5	TYPE D120
3-20†	100-300	1-10	5-25
5-50†		5-25	10-75
10-100			40-150
20-150			

†With Vacuum Spring Assembly, minimum range is 30 inches Hg; maximum is reduced by 15 PSIG.

Canadian Registration # OC 0591.9C

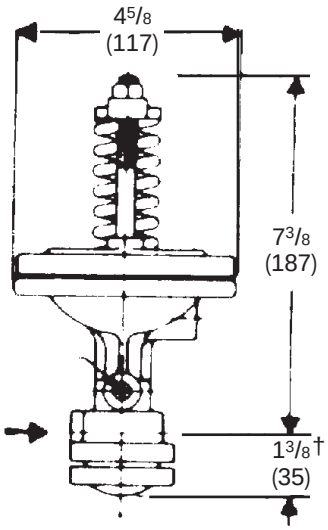
TYPE D
PRESSURE REDUCING PILOT

SPECIFICATION

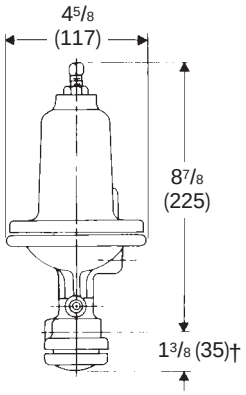
The Pilot shall be separate from the main valve and connected to it with a male union. The Pilot shall be normally closed design with packless construction. A strainer screen shall be built into the Pilot inlet. The Pilot shall be interchangeable on all sizes of main valves.

MATERIALS OF CONSTRUCTION

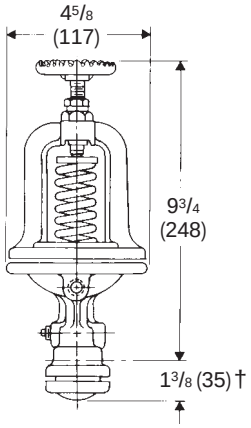
Body, Cast Iron .....ASTM A126 Cl B
Body, Cast Steel .....ASTM A216 GR. WCB
Stem .....303 St. Stl. ASTM A582 COND A
Disc .....440 St. St. ASTM A276-75 COND A
Seat .....420 St. Stl ASTM A276 COND A
Gasket .....Non-Asbestos
Diaphragm .....301 St. Stl. MIL-5-5059C
Spring .....Inconel



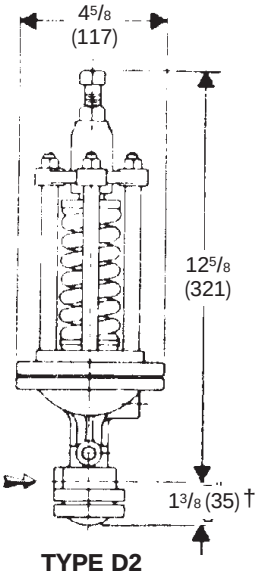
STANDARD D PILOT
7 LBS.
(3.2 KG)



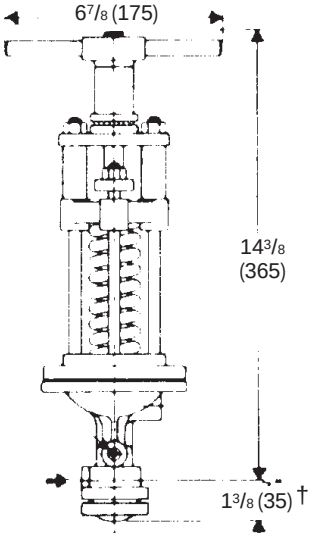
D SPRING CHAMBER
8 LBS. (3.6 KG)



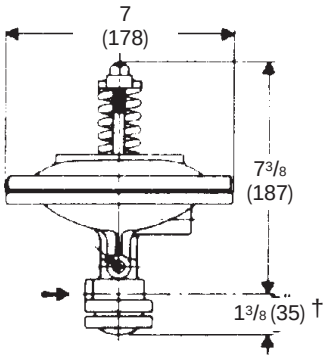
D ADJUSTING
HANDWHEEL
9 LBS. (4.1 KG)



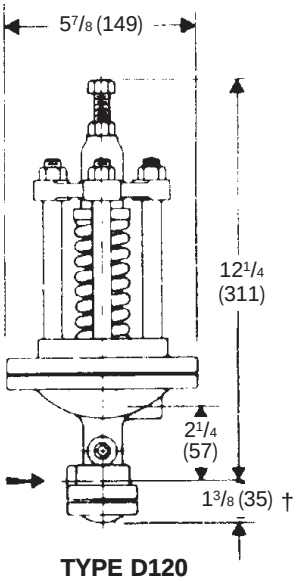
TYPE D2
10 LBS.
(4.5 KG)



D2 ADJUSTING
HANDWHEEL
12 LBS. (5.4 KG)



TYPE D5
14 LBS.
(6.4 KG)



TYPE D120
16 LBS.
(7.3 KG)

†For Integral Mount Pilot, this dimension is 5/8" (16).